

Section : Chemistry

Q.1 Arrange the following compounds in the decreasing order of their boiling points:

- (A) Butyraldehyde
- (B) n-Butyl alcohol
- (C) Sec-butyl alcohol
- (D) n-butane

Choose the correct answer from the options given below:

- 1. (A), (B), (D), (C)
- 2. (A), (B), (C), (D)
- 3. (B), (C), (A), (D)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.2 The conductivity of a 0.001 mol L^{-1} solution of a weak monobasic acid is $4.0 \times 10^{-5} \text{ S cm}^{-1}$.

Calculate its dissociation constant if Λ_m° for the weak acid is $400.0 \text{ S cm}^2 \text{ mol}^{-1}$.

- 1. $1.5 \times 10^{-4} \text{ mol L}^{-1}$
- 2. $1.1 \times 10^{-4} \text{ mol L}^{-1}$
- 3. $1.5 \times 10^{-5} \text{ mol L}^{-1}$
- 4. $1.1 \times 10^{-5} \text{ mol L}^{-1}$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.3 A first order reaction is found to have rate constant $k = 3.0 \times 10^{-1} \text{ s}^{-1}$. The half-life of the reaction will be:

- 1. 0.650 s
- 2. 0.208 s
- 3. 2.31 s
- 4. 0.993 s

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.4 The rate constant of a reaction increases from 0.02 s^{-1} to 0.04 s^{-1} when the temperature is raised from 500 K to 600 K. The activation energy (E_a) of the reaction is

(Given: $\log_{10} 2 = 0.3010$)

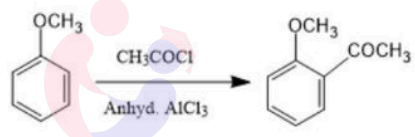
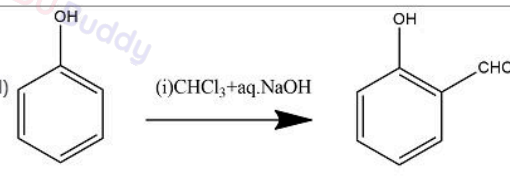
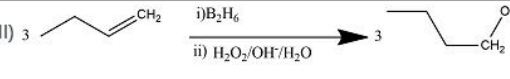
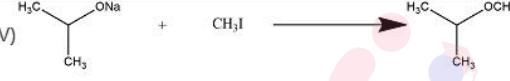
- 1. $0.170 \text{ cal mol}^{-1}$
- 2. $17.289 \text{ kJ mol}^{-1}$
- 3. $7.513 \text{ cal mol}^{-1}$
- 4. $7.521 \text{ kJ mol}^{-1}$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.5

Match List-I with List-II

List-I	List-II
(Name reaction.)	(Reaction)
(A) Williamson ether synthesis	(I) 
(B) Hydroboration oxidation	(II) 
(C) Reimer-Tiemann reaction	(III) 
(D) Friedel Craft Acylation reaction	(IV) 

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Mischmetall consists of

- Lanthanoid metal (~95%), Iron (~5%), traces of S, C, Ca and Al
- Lanthanoid metal (~85%), Iron (~15%), traces of S, C, Ca and Al
- Actinoid metal (~90%), Iron (~10%), traces of S, C, Ca and Al
- Actinoid metal (~95%), Iron (~3%), traces of C, Ca and A

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Chloroform is largely used for

- Production of freon
- As an antiseptic
- Propellant for aerosol cans
- Paint remover

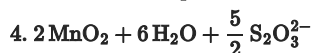
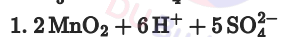
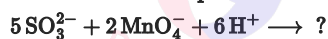
Options 1. 1

2. 2

3. 3

4. 4

Q.8 Choose the correct products formed in acidic medium.



Options 1. 1

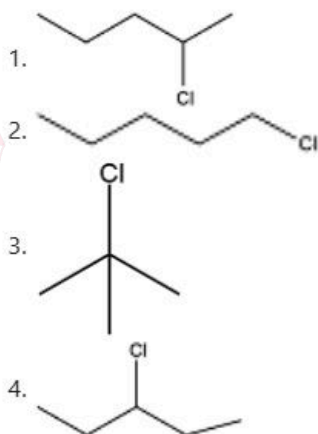
2. 2

3. 3

4. 4

Q.9

Which of the following compounds will undergo S_N1 reaction at a fastest rate?



Options 1. 1

2. 2

3. 3

4. 4

Q.10 Match List-I with List-II

List-I

(Types of solutions)

List-II

(Examples)

(A) Gas in Gas solution

(I) Oxygen dissolved in water

(B) Gas in Liquid solution

(II) Copper dissolved in gold

(C) Liquid in Solid solution (III) Mixture of oxygen and nitrogen gases

(D) Solid in Solid solution (IV) Amalgam of mercury with sodium

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

4. (A) - (I), (B) - (IV), (C) - (III), (D) - (II)

Options 1. 1

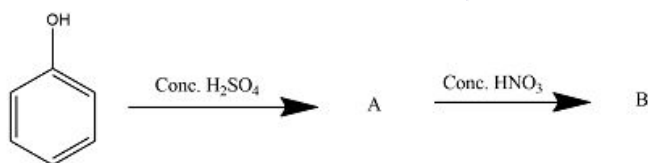
2. 2

3. 3

4. 4

Q.11

Consider the following reaction and identify A and B.



1. A: Phenol-2,4-disulphonic acid; B: Picric acid

2. A: Phenol-4-sulphonic acid; B: 4-Nitrophenol

3. A: Phenol-2,6- disulphonic acid; B: 2, 5-dinitrophenol

4. A: Phenol-4-sulphonic acid; B: 2,4,6-trinitrophenol

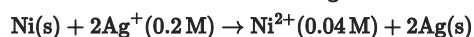
Options 1. 1

2. 2

3. 3

4. 4

Q.12 Calculate the emf of the following cell at 298 K if $E^\circ_{\text{Cell}} = 1.10 \text{ V}$:



1. 1.10 V

2. 1.07 V

3. 1.05 V

4. 0.030 V

Options 1. 1

2. 2

3. 3

4. 4

Q.13 The characteristic oxidation state of Lanthanoids is +3, but Ce shows +4 oxidation state because...

(A) It has variable ionization enthalpy

(B) It has a tendency to attain noble gas configuration

(C) It has a tendency to attain f^0 configuration

(D) It resembles Pb^{4+}

Choose the correct answer from the options given below:

1. (A), (B) and (C) only

2. (C) and (D) only

3. (A) and (C) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Minimum boiling azeotropes show deviation from Raoult's Law at a specific composition.

1. Small Positive

2. Small Negative

3. Large Positive

4. Large Negative

Options 1. 1

2. 2

3. 3

4. 4

Q.15 An organic compound (A), $C_9H_{10}O$, gives a 2,4-DNP reaction, negative Tollens' test and iodoform test. On drastic oxidation with chromic acid, it forms B, having molecular formulae, $C_7H_6O_2$. Identify A and B.

1. A: 3-Phenylpropanal, B: Benzoic acid

2. A: Propiophenone, B: Benzoic acid

3. A: 1-Phenylacetophenone, B: Benzoic acid

4. A: 3-Phenylpropanal, B: Phenylacetic acid

Options 1. 1

2. 2

3. 3

4. 4

Q.16 The resistance of 1N solution of four different electrolytes are given below. Arrange them in increasing order of their equivalent conductance values if the same conductivity cell was used to measure the resistance of all the electrolytic solutions:

(A) 250 ohm

(B) 150 ohm

(C) 50 ohm

(D) 280 ohm

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (B), (D), (C)

3. (D), (A), (B), (C)

4. (C), (B), (A), (D)

Options 1. 1

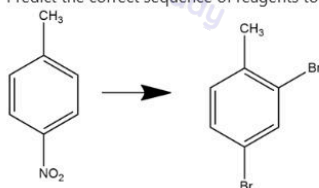
2. 2

3. 3

4. 4

Q.17

Predict the correct sequence of reagents to carry out the following conversion:



1. (i) Cu_2Br_2/HBr (ii) Sn/HCl (iii) $NaNO_2/HCl$ (iv) $Br_2/FeBr_3$

2. (i) $Br_2/FeBr_3$ (ii) Sn/HCl (iii) $NaNO_2/HCl$ (iv) Cu_2Br_2/HBr

3. (i) $Br_2/FeBr_3$ (ii) Sn/HCl (iii) Cu_2Br_2/HBr (iv) $NaNO_2/HCl$

4. (i) Sn/HCl (ii) $Br_2/FeBr_3$ (iii) $NaNO_2/HCl$ (iv) Cu_2Br_2/HBr

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Nitrogen in trimethylamine is sp^3 hybridized. The C-N-C bond angle is 108° . The geometry of trimethylamine is

1. Tetrahedral
2. Trigonal bipyramidal
3. Square planar
4. Pyramidal

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Given below are four statements with reference to a catalyst. Choose the incorrect statement(s)

- (A) A catalyst lowers the potential energy barrier, thus decreasing the activation energy.
(B) A catalyst lowers the Gibbs energy (ΔG) of the reaction.
(C) A small amount of catalyst converts a large amount of reactants into products.
(D) A catalyst can catalyze non-spontaneous reactions also.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B) and (D) only
3. (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.20 The increasing order of the melting points of following transition metals is

- (A) Fe
(B) Mn
(C) Cr
(D) V

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Carbonyl carbon in acetone is sp^2 hybridized. The C-O-C bond angle and the geometry of acetone is.....

1. 120° ; trigonal coplanar
2. 180° ; linear
3. $109^\circ 28'$; Tetrahedral
4. 60° ; trigonal planar

Options 1. 1

2. 2

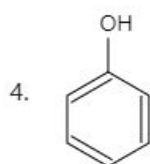
3. 3

4. 4

Q.22

Pick the correct structure for carboic acid among the following compounds:

1. CH_3OH
2. CH_3COOH
3. H_2CO_3



Options 1. 1

2. 2

3. 3

Q.23 Edema is caused by

1. Water retention (osmosis) due to high intake of salt
2. Water loss (reverse osmosis) due to high intake of salt
3. Water retention (reverse osmosis) due to low intake of salt
4. Water loss (osmosis) due to high intake of salt

Options 1. 1

2. 2
3. 3
4. 4

Q.24

Match List-I with List-II

List-I	List-II
(Reaction)	(Name)
(A) $\text{CH}_3\text{CHO} + \text{Dil NaOH} \longrightarrow \text{CH}_3\text{CHOHCH}_2\text{CHO}$	(I) Aldol Reaction
(B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{Br}_2 \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) Red P}} \text{CH}_3\text{CH}_2\text{CHBrCOOH}$	(II) Cannizzaro Reaction
(C) $\text{C}_6\text{H}_5\text{CHO} + \text{Conc. KOH} \longrightarrow \text{C}_6\text{H}_5\text{COONa} + \text{C}_6\text{H}_5\text{CH}_2\text{OH}$	(III) Haloform Reaction
(D) $\text{CH}_3\text{COCH}_3 \xrightarrow{\text{NaOCl}} \text{CH}_3\text{COONa} + \text{CHCl}_3$	(IV) Hell Volhard Zelinski Reaction

Choose the **correct** answer from the options given below:

1. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Arrange the following compounds in the decreasing order of their reactivity towards Nucleophilic substitution reaction:

- (A) Bromobenzene
- (B) 4-Nitrobromobenzene
- (C) 2,4,6-Trinitrobromobenzene
- (D) 2,4-Dinitrobromobenzene

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (D), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Predict the percentage of nitration products when aniline is subjected

to nitration with conc. HNO_3 /conc. H_2SO_4 .

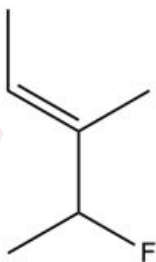
1. *o*-nitroaniline 51%, *m*-nitroaniline 47%, *p*-nitroaniline 2%
2. *o*-nitroaniline 2%, *m*-nitroaniline 47%, *p*-nitroaniline 51%
3. *o*-nitroaniline 47%, *m*-nitroaniline 51%, *p*-nitroaniline 2%
4. *o*-nitroaniline 51%, *m*-nitroaniline 2%, *p*-nitroaniline 47%

Options 1. 1

2. 2
3. 3
4. 4

Q.27

Choose the correct IUPAC name of the following compound:



1. 3-Methyl-4-fluoropent-2-ene
2. 4-Fluoro-3-methylpent-2-ene
3. 3-Fluoro-3-methylpent-3-ene
4. 4-Fluoro-3-methylpent-3-ene

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Arrange the following compounds in increasing order of their acidic character.

- (A) m-nitrophenol
- (B) p-nitrophenol
- (C) Phenol
- (D) p-methoxyphenol

Choose the correct answer from the options given below:

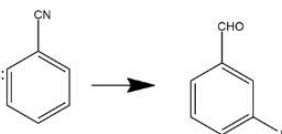
1. (D), (B), (C), (A)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (D), (C), (A), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.29

The correct sequence of reagents to carry out the following conversion is:



1. (i) Sn/HCl (ii) Conc.HNO₃/Conc.H₂SO₄ (iii) DIBAL-H (iv) NaNO₂/HCl (v) HBF₄
2. (i) Conc.HNO₃/Conc.H₂SO₄ (ii) Sn/HCl (iii) DIBAL-H (iv) NaNO₂/HCl (v) HBF₄
3. (i) NaNO₂/HCl (ii) DIBAL-H (iii) Sn/HCl (iv) Conc.HNO₃/Conc.H₂SO₄ (v) HBF₄
4. (i) NaNO₂/HCl (ii) DIBAL-H (iii) Sn/HCl (iv) HBF₄ (v) Conc.HNO₃/Conc.H₂SO₄

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Which of the following will not give iodoform reaction?

1. Acetaldehyde
2. Isopropyl alcohol
3. Butanone
4. Propionaldehyde

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The mass percentage of a component of a solution is defined as:

1. $\frac{\text{Mass of component in the solution}}{\text{Total mass of solvent}} \times 100$

2. $\frac{\text{Mass of component in the solution}}{\text{Total mass of solution}} \times 100$
3. $\frac{\text{Mass of component in the solution}}{\text{Total volume of solvent}} \times 100$
4. $\frac{\text{Mass of component in the solution}}{\text{Total volume of solution}} \times 100$

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Λ_m° for NaCl, HCl, and CH_3COONa are 126.4, 425.9, and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$, respectively. Calculate Λ_m° for CH_3COOH .

1. $461.3 \text{ S cm}^2 \text{ mol}^{-1}$
2. $425.9 \text{ S cm}^2 \text{ mol}^{-1}$
3. $308.5 \text{ S cm}^2 \text{ mol}^{-1}$
4. $390.5 \text{ S cm}^2 \text{ mol}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which of the following compound is a gem-dihalide?

1. 1,2-Dichloroethane
2. Ethylene dichloride
3. Methylene chloride
4. Benzyl chloride

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Match List-I with List-II

List-I	List-II
Reagent	Characteristic of Compound
(A) Tollen's Reagent	(I) Aliphatic aldehyde
(B) Iodoform test	(II) Tertiary alcohol
(C) Lucas Reagent	(III) Methyl ketone
(D) Fehling's Reagent	(IV) Aliphatic and aromatic aldehydes

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.35 The rate constant for a reaction is $3.5 \text{ mol}^{-1} \text{ L s}^{-1}$. The order of the reaction is:

1. Zero
2. First
3. Second
4. Third

Options 1. 1

2. 2
3. 3
4. 4

Q.36 The limiting molar conductivities for H^+ , Na^+ , Cl^- , CH_3COO^- are 349.6, 50.1, 76.3, $40.9 \text{ S cm}^2 \text{ mol}^{-1}$, respectively, in water at 298 K.

Match List-I and List-II

List-I	LIST-II
Electrolyte	Λ_m°
(A) HCl	(I) $390.5 \text{ S cm}^2 \text{ mol}^{-1}$
(B) NaCl	(II) $126.4 \text{ S cm}^2 \text{ mol}^{-1}$
(C) CH_3COONa	(III) $425.9 \text{ S cm}^2 \text{ mol}^{-1}$

(D) CH_3COOH (IV) $91.0 \text{ S cm}^2 \text{ mol}^{-1}$

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Which of the following statements is/ are correct

- (A) For an ideal solution, the enthalpy and volume of mixing of pure components are zero.
- (B) The inter-molecular attractive forces between A-A and B-B are greater than those between A-B for an ideal binary solution.
- (C) The inter-molecular attractive forces between A-A and B-B are of the same order as those between A-B for an ideal binary solution
- (D) For an ideal solution, the enthalpy and volume of mixing of pure components to form the solution are positive.

Choose the correct answer from the options given below:

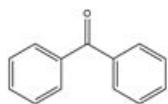
1. (C) and (D) only
2. (A), (B) and (D) only
3. (A) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.38

The common name of the following compound is



1. Acetophenone
2. Benzophenone
3. Propiophenone
4. Anisole

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Which of the following statements is not correct?

1. Mn^{2+} and Co^{3+} are oxidizing agents in aqueous solution
2. Ti^{2+} and Cr^{2+} are reducing agents in aqueous solution
3. Cu liberates hydrogen from acids
4. in its higher oxidation states, Mn forms stable compounds with oxygen and fluorine.

Options 1. 1

2. 2
3. 3
4. 4

Q.40 Which of the following statements are true?

- (A) Decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction.
- (B) The rate of a reaction increases with the passage of time.
- (C) A catalyst does not change the equilibrium of a reaction.
- (D) The molecularity of a reaction can be zero, fraction and negative.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A) and (C) only
3. (B) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of dipolar molecules like NH₃ and H₂O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

The hybridisation and magnetism of the complexes $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{Mn}(\text{CN})_6]^{3-}$ respectively is:

1. (d^2sp^3 , diamagnetic) and (d^2sp^3 , diamagnetic)
2. (sp^3d^2 , diamagnetic) and (d^2sp^3 , paramagnetic)
3. (d^2sp^3 , diamagnetic) and (d^2sp^3 , paramagnetic)
4. (d^2sp^3 , paramagnetic) and (sp^3d^2 , diamagnetic)

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the passage carefully and answer

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of dipolar molecules like NH₃ and H₂O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

The colour of coordination compounds is attributed to

1. s-p transition
2. d-d transition
3. d-f transition
4. p-p* transition

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Read the passage carefully and answer

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of dipolar molecules like NH₃ and H₂O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

If the fourth electron in d subshell enters one of the e_g orbitals in octahedral complexes, it indicates that

1. the ligand is weak, and forms low spin complexes.
2. the ligand is strong, and forms low spin complexes.
3. the ligand is strong, and forms high spin complexes.
4. the ligand is weak, and forms high spin complexes.

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of dipolar molecules like NH₃ and H₂O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

In tetrahedral coordination complexes, low-spin configurations are rare because:

1. d-orbital splitting in tetrahedral complexes causes more distortion as compared to octahedral complexes.
2. d-orbital splitting is inverted as compared to octahedral complexes.

$$3. \Delta_t = \frac{9}{4} \Delta_o$$

4. Δ_t is not sufficiently large to cause pairing of electrons.

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of dipolar molecules like NH_3 and H_2O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

Which of the following pairs exists as high-spin, tetrahedral complexes?

1. $[\text{Ni}(\text{SCN})_4]^{2-}$ and $[\text{Mn}(\text{CN})_4]^{2-}$
2. $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{MnBr}_4]^{2-}$
3. $[\text{NiCl}_4]^{2-}$ and $[\text{MnBr}_4]^{2-}$
4. $[\text{NiF}_4]^{2-}$ and $[\text{Mn}(\text{CN})_4]^{2-}$

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer

Carbohydrates are essential for both plants and animals. Some common examples of carbohydrates are glucose, fructose, sucrose, glycogen, cellulose and starch. They form a major portion of our food. Honey has been used for a long time as an instant source of energy in Ayurvedic medicine. Carbohydrates are used as storage molecules in both plants and animals. The cell wall of bacteria and plants is made of cellulose. We use cellulose in the form of wood for making furniture and as cotton fiber for clothing. They also provide raw material for textiles, paper, lacquers, and breweries. Carbohydrates are classified as Monosaccharides, Oligosaccharides and Polysaccharides on the basis of their behavior towards hydrolysis. Monosaccharides cannot be hydrolyzed further to simpler units whereas oligosaccharides can yield 2-10 units of monosaccharides on hydrolysis. Polysaccharides, which are non-sugars yield a large number of monosaccharide units on hydrolysis.

Answer the following questions about carbohydrates:

Identify a component of starch from among the following.

1. Amylase
2. Amylose
3. Galactose
4. Fructose

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer

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Answer the following questions about carbohydrates:

Pick the incorrect statement from the following:

1. Amylopectin contains only 1,3-glycosidic linkage
2. Amylopectin is insoluble in water
3. Amylopectin is the major constituent of starch
4. Amylopectin has a branched chain structure.

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer

Carbohydrates are essential for both plants and animals. Some common examples of carbohydrates are glucose, fructose, sucrose, glycogen, cellulose and starch. They form a major portion of our food. Honey has been used for a long time as an instant source of energy in Ayurvedic medicine. Carbohydrates are used as storage molecules in both plants and animals. The cell wall of bacteria and plants is made of cellulose. We use cellulose in the form of wood for making furniture and as cotton fiber for clothing. They also provide raw material for textiles,

paper, lacquers, and breweries. Carbohydrates are classified as Monosaccharides, Oligosaccharides and Polysaccharides on the basis of their behavior towards hydrolysis. Monosaccharides cannot be hydrolyzed further to simpler units whereas oligosaccharides can yield 2-10 units of monosaccharides on hydrolysis. Polysaccharides, which are non-sugars yield a large number of monosaccharide units on hydrolysis.

Answer the following questions about carbohydrates:

Hydrolytic products of maltose are?

1. Glucose and glucose
2. Glucose and fructose
3. Glucose and galactose
4. Galactose and fructose

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer

Carbohydrates are essential for both plants and animals. Some common examples of carbohydrates are glucose, fructose, sucrose, glycogen, cellulose and starch. They form a major portion of our food. Honey has been used for a long time as an instant source of energy in Ayurvedic medicine. Carbohydrates are used as storage molecules in both plants and animals. The cell wall of bacteria and plants is made of cellulose. We use cellulose in the form of wood for making furniture and as cotton fiber for clothing. They also provide raw material for textiles, paper, lacquers, and breweries. Carbohydrates are classified as Monosaccharides, Oligosaccharides and Polysaccharides on the basis of their behavior towards hydrolysis. Monosaccharides cannot be hydrolyzed further to simpler units whereas oligosaccharides can yield 2-10 units of monosaccharides on hydrolysis. Polysaccharides, which are non-sugars yield a large number of monosaccharide units on hydrolysis.

Answer the following questions about carbohydrates:

Animal starch is popularly known as

1. Starch
2. Cellulose
3. Amylose
4. Glycogen

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer

Carbohydrates are essential for both plants and animals. Some common examples of carbohydrates are glucose, fructose, sucrose, glycogen, cellulose and starch. They form a major portion of our food. Honey has been used for a long time as an instant source of energy in Ayurvedic medicine. Carbohydrates are used as storage molecules in both plants and animals. The cell wall of bacteria and plants is made of cellulose. We use cellulose in the form of wood for making furniture and as cotton fiber for clothing. They also provide raw material for textiles, paper, lacquers, and breweries. Carbohydrates are classified as Monosaccharides, Oligosaccharides and Polysaccharides on the basis of their behavior towards hydrolysis. Monosaccharides cannot be hydrolyzed further to simpler units whereas oligosaccharides can yield 2-10 units of monosaccharides on hydrolysis. Polysaccharides, which are non-sugars yield a large number of monosaccharide units on hydrolysis.

Answer the following questions about carbohydrates:

The sugar having β -glycosidic linkage is

1. Starch
2. Glycogen
3. Cellulose
4. Maltose

Options 1. 1

2. 2
3. 3
4. 4